Test Plan Document: Swag Labs (SauceDemo) E-Commerce Platform

Project: Swag Labs QA Testing

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Target Application: https://www.saucedemo.com

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1.Scope

1.1 In-Scope Features

- **User Authentication:** Login, Logout, Cookie/Session retention, and error handling for restricted users.
- **Product Catalog:** Product listing display, sorting (A-Z, Z-A, Price Low-High, Price High-Low), image rendering, and individual product detail view.
- **Shopping Cart:** Adding/Removing items from catalog & detail view, cart badge counter updates, and cart item persistence.
- **Checkout Workflow:** Step 1 Information input (First Name, Last Name, Postal Code), Step 2 Overview (item verification, tax & total calculation), and Step 3 Order Completion.

1.2 Out-of-Scope

- Payment gateway integrations (stumped/simulated UI only).

2. Types of Testing

1. **Functional Testing:** Verification of core flows like authentication, product selection, and checkout validation.
2. **UI / Usability Testing:** Visual layout checking, responsiveness, element alignment, and broken image detection.
3. **Negative Testing:** Submitting invalid credentials, blank inputs during checkout, or performing unauthorized actions.
4. **Edge Cases:** Empty cart checkout attempts, extreme character inputs in checkout fields, sorting state continuity.
5. **Cross-Browser & Compatibility Testing:** Validating performance and rendering across Google Chrome, Mozilla Firefox, and Apple Safari

3. Test Environment

- **Operating System:** Windows 11
- **Web Browsers:** Google Chrome (Latest)
- **Screen Resolutions:** Desktop (1920x1080), Laptop (1366x768), Mobile Viewport (Responsive 375x812)
- **Automation Tools:** Python 3.x with Selenium WebDriver & PyTest

4. Test Data

- **Base URL:** https://www.saucedemo.com
- **Default Password (All Users):** `secret_sauce`
- **User Profiles:**

- **standard_user**: The normal functional user. It logs in successfully with no intentional bugs or performance issues.
- **locked_out_user**: Triggers a login restriction error immediately upon attempting to log in.
- **problem_user**: Contains deliberate functional and UI bugs (such as broken product images, inaccurate filtering, and broken form elements).
- **performance_glitch_user**: Simulates high latency and slow network loading times during login and page navigation.
- **error_user**: A user profile that intentionally triggers various application errors during specific user flows (such as updating cart items or checkout actions).
- **visual_user**: A user profile specifically designed to test visual regressions, where layout elements, alignments, or styling might render differently or incorrectly compared to the standard view.

6. Detailed Test Cases (10 Test Cases)

Test Case ID	Feature	Precondition	Objective	Priority	Regression	Test Data	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-01	Authentication	Browser is navigated to https://www.saucedemo.com	To verify that a valid user can successfully log in and access the inventory dashboard.	High	Yes	Username: standard_user Password: secret_sauce	1. Enter standard_user in the Username field. 2. Enter secret_sauce in the Password field. 3. Click the "Login" button.	User is successfully redirected to inventory.html and header displays "Swag Labs".	User was redirected to the inventory page successfully.	Pass
TC-02	Authentication / Negative Testing	User is on the Login Page	To verify that a locked-out user is blocked from	High	Yes	Username: locked_out_user Password: secret_sauce	1. Enter locked_out_user in Username. 2. Enter secret_sau	Login fails and error banner appea	Error banner displaye	Pass

			logging in and receives an appropriate error message.				ce in Password. 3. Click "Login".	rs: *"Epic sadface: Sorry, this user has been locked out."*	d correctly as expected.	
TC-03	Product Catalog	User is logged in as standard_user and on the Inventory page	To verify that products can be correctly sorted by price from low to high.	Medium	Yes	Selected Filter: Price (low to high)	1. Click on the sort dropdown menu (top right corner). 2. Select "Price (low to high)".	Products are re-ordered sequentially with prices ascending from lowest to highest.	Products sorted successfully in ascending order.	Pass
TC-04	Cart & Checkout Flow	User is logged in as standard_user	To verify the complete end-to-end shopping cart and checkout order placement workflow.	High	Yes	Item: Sauce Labs Backpack First Name: John Last Name: Doe Postal Code: 10001	1. Add "Sauce Labs Backpack" to cart. 2. Go to Cart and click "Checkout". 3. Fill in First Name, Last Name, and Zip Code.	Order is completed successfully and header displays *"Thank you for your order!"*	Order confirmation screen rendered successfully.	Pass

							4. Click "Continue", review summary, and click "Finish".			
TC-05	Checkout Information Validation / Negative Testing	User has added an item to cart and is on checkout-step-one.html	To verify that the system blocks checkout progression when mandatory info fields are empty.	High	Yes	First Name: *(Blank)* Last Name: *(Blank)* Zip Code: *(Blank)*	1. Leave First Name, Last Name, and Postal Code empty. 2. Click the "Continue" button.	Navigation is blocked and error message appears: *"Error: First Name is required"*.	Error message triggered successfully, blocking progression.	Pass
TC-06	Authentication / Negative Testing	User is on the Login Page [https://www.saucedemo.com]	To verify that entering invalid credentials blocks login and displays an appropriate error message	High	Yes	Username: invalid_user, Password: wrong_password	1.Enter invalid_user in the Username field. 2. Enter wrong_password in the Password field. 3.Click the "Login" button	Login fails, navigation is blocked, and an error message appears stating: *"Epic sadface: Username and password*	System blocked access and displayed the incorrect credentials error banner.	Pass

								do not match any user in this service".		
TC-07	Authentication / Negative Testing	User is on the Login Page(https://www.saucedemo.com)	o verify that attempting to log in with empty username/ password fields triggers a mandatory field validation error.	High	Yes	Username: *(Blank)*, Password: *(Blank)*	1.Leave the Username and Password fields completely blank. 2.Click the "Login" button.	Login fails and an error message appears stating: *"Epic sadface: Username is required".*	Login fails and an error message appears stating: *"Epic sadface: Username is required".*	Pass
TC-08	Shopping Cart	User is logged in as standard_user and on the Inventory page (inventory.html)	To verify that clicking "Add to cart" updates the cart button state and increments the shopping	High	Yes	Item Name: Sauce Labs Backpack	1.Locate "Sauce Labs Backpack" on the inventory page. 2.Click the "Add to cart" button.	The button text changes to "Remove" and the shopping cart badge	The button text changes to "Remove" and the sho	Pass

			cart badge counter					icon at the top right displays 1	ppi ng cart bad ge icon at the top righ t disp lays 1	
TC -09	Shoppi ng Cart	User is logged in as standard_u ser, has added "Sauce Labs Backpack" to the cart, and is viewing the Cart page (cart.html)	To verify that removing an item from the cart updates the item list and removes the cart badge counter	Hi gh	Ye s	Item Name: Sauce Labs Backpack	1.Click on the Shopping Cart icon at the top right. 2.Click the "Remove" button next to "Sauce Labs Backpack"	The item is remo ved from the cart list view and the cart badge count er disap pears/ resets	The ite m is rem ove d fro m the cart list vie w and the cart bad ge cou nter disa ppe ars/ rese ts	Pas s
TC -10	Checko ut Workfl ow	User is logged in as standard_u ser, has at least 1 item in the cart, and is on the	To verify that clicking the "Checkout " button successfull y redirects the user to	Hi gh	Ye s	Cart Items: 1 item added	1.Click on the Shopping Cart icon to view the cart. 2. Click the	User is redire cted to (https ://ww w.sau cede mo.co	Use r was suc cess full y redi rect	Pas s

		Cart page (cart.html)	the Checkout Informatio n page				"Checkout " button.	m/che ckout -step- one.ht ml) displa ying the "Chec kout: Your Infor matio n" title.	ed to the che cko ut step one pag e.	

Test Case 1

- **Test Case ID:** TC- 01
- **Feature:** Authentication
- **Precondition:** Browser is navigated to (https://www.saucedemo.com)
- **Objective:** To verify that a valid user can successfully log in and access the inventory dashboard.
- **Priority:** High
- **Regression:** Yes
- **Test Data:** Username: standard_user, Password: secret_sauce
- **Test Steps:**
    1. Enter standard_user in the Username field.
    2. Enter secret_sauce in the Password field.
    3. Click the "Login" button.
- **Expected Result:** User is successfully redirected to inventory.html and header displays "Swag Labs".
- **Actual Result:** User was redirected to the inventory page successfully.
- **Pass/Fail:** Pass

Test Case 2

- **Test Case ID:** TC-02
- **Feature:** Authentication / Negative Testing
- **Precondition:** User is on the Login Page
- **Objective:** To verify that a locked-out user is blocked from logging in and receives an appropriate error message.
- **Priority:** High
- **Regression:** Yes
- **Test Data:** Username: locked_out_user, Password: secret_sauce
- **Test Steps:**
    1. Enter locked_out_user in Username.
    2. Enter secret_sauce in Password.
    3. Click "Login".

- **Expected Result:** Login fails and error banner appears: *"Epic sadface: Sorry, this user has been locked out."*
- **Actual Result:** Error banner displayed correctly as expected.
- **Pass/Fail:** Pass

Test Case 3

- **Test Case ID:** TC-03
- **Feature:** Product Catalog
- **Precondition:** User is logged in as standard_user and on the Inventory page
- **Objective:** To verify that products can be correctly sorted by price from low to high.
- **Priority:** Medium
- **Regression:** Yes
- **Test Data:** Selected Filter: Price (low to high)
- **Test Steps:**
    1. Click on the sort dropdown menu (top right corner).
    2. Select "Price (low to high)".
- **Expected Result:** Products are re-ordered sequentially with prices ascending from lowest to highest.
- **Actual Result:** Products sorted successfully in ascending order.
- **Pass/Fail:** Pass

Test Case 4

- **Test Case ID:**TC-04
- **Feature:** Cart & Checkout Flow
- **Precondition:** User is logged in as standard_user
- **Objective:** To verify the complete end-to-end shopping cart and checkout order placement workflow.
- **Priority:** High
- **Regression:** Yes
- **Test Data:** Item: Sauce Labs Backpack, First Name: John, Last Name: Doe, Postal Code: 10001
- **Test Steps:**
    1. Add "Sauce Labs Backpack" to cart.
    2. Go to Cart and click "Checkout".
    3. Fill in First Name, Last Name, and Zip Code.
    4. Click "Continue", review summary, and click "Finish".
- **Expected Result:** Order is completed successfully and header displays *"Thank you for your order!"*
- **Actual Result:** Order confirmation screen rendered successfully.
- **Pass/Fail:** Pass

Test Case 5

- **Test Case ID:** TC-05
- **Feature:** Checkout Information Validation / Negative Testing
- **Precondition:** User has added an item to cart and is on checkout-step-one.html
- **Objective:** To verify that the system blocks checkout progression when mandatory info fields are empty.

- **Priority:** High
- **Regression:** Yes
- **Test Data:** First Name: *(Blank)*, Last Name: *(Blank)*, Zip Code: *(Blank)*
- **Test Steps:**
    1. Leave First Name, Last Name, and Postal Code empty.
    2. Click the "Continue" button.
- **Expected Result:** Navigation is blocked and error message appears: *"Error: First Name is required"*.
- **Actual Result:** Error message triggered successfully, blocking progression.
- **Pass/Fail:** Pass

Test Case 6

- **Test Case ID:** TC-06
- **Feature:** Authentication / Negative Testing
- **Precondition:** User is on the Login Page (https://www.saucedemo.com)
- **Objective:** To verify that entering invalid credentials blocks login and displays an appropriate error message.
- **Priority:** High
- **Regression:** Yes
- **Test Data:** Username: invalid_user, Password: wrong_password
- **Test Steps:**
    1. Enter invalid_user in the Username field.
    2. Enter wrong_password in the Password field.
    3. Click the "Login" button.
- **Expected Result:** Login fails, navigation is blocked, and an error message appears stating: *"Epic sadface: Username and password do not match any user in this service"*.
- **Actual Result:** System blocked access and displayed the incorrect credentials error banner.
- **Pass/Fail:** Pass

Test Case 7

- **Test Case ID:** TC-07
- **Feature:** Authentication / Negative Testing
- **Precondition:** User is on the Login Page (https://www.saucedemo.com)
- **Objective:** To verify that attempting to log in with empty username/password fields triggers a mandatory field validation error.
- **Priority:** High
- **Regression:** Yes
- **Test Data:** Username: *(Blank)*, Password: *(Blank)*
- **Test Steps:**
    1. Leave the Username and Password fields completely blank.
    2. Click the "Login" button.
- **Expected Result:** Login fails and an error message appears stating: *"Epic sadface: Username is required"*.
- **Actual Result:** System blocked submission and displayed the required username error message.
- **Pass/Fail:** Pass

Test Case 8

- **Test Case ID:** TC-08
- **Feature:** Shopping Cart
- **Precondition:** User is logged in as standard_user and on the Inventory page (inventory.html).
- **Objective:** To verify that clicking "Add to cart" updates the cart button state and increments the shopping cart badge counter.
- **Priority:** High
- **Regression:** Yes
- **Test Data:** Item Name: Sauce Labs Backpack
- **Test Steps:**
    1. Locate "Sauce Labs Backpack" on the inventory page.
    2. Click the "Add to cart" button.
- **Expected Result:** The button text changes to "Remove" and the shopping cart badge icon at the top right displays 1.
- **Actual Result:** Cart badge updated to 1 and button label changed to "Remove".
- **Pass/Fail:** Pass

Test Case 9

- **Test Case ID:** TC_CART_02
- **Feature:** Shopping Cart
- **Precondition:** User is logged in as standard_user, has added "Sauce Labs Backpack" to the cart, and is viewing the Cart page (cart.html).
- **Objective:** To verify that removing an item from the cart updates the item list and removes the cart badge counter.
- **Priority:** High
- **Regression:** Yes
- **Test Data:** Item Name: Sauce Labs Backpack
- **Test Steps:**
    1. Click on the Shopping Cart icon at the top right.
    2. Click the "Remove" button next to "Sauce Labs Backpack".
- **Expected Result:** The item is removed from the cart list view and the cart badge counter disappears/resets.
- **Actual Result:** Item was removed from the list and the cart badge counter disappeared.
- **Pass/Fail:** Pass

Test Case 10

- **Test Case ID:** TC-10
- **Feature:** Checkout Workflow
- **Precondition:** User is logged in as standard_user, has at least 1 item in the cart, and is on the Cart page (cart.html).
- **Objective:** To verify that clicking the "Checkout" button successfully redirects the user to the Checkout Information page.
- **Priority:** High
- **Regression:** Yes
- **Test Data:** Cart Items: 1 item added

- **Test Steps:**
    1. Click on the Shopping Cart icon to view the cart.
    2. Click the "Checkout" button.
- **Expected Result:** User is redirected to (https://www.saucedemo.com/checkout-step-one.html) displaying the "Checkout: Your Information" title.
- **Actual Result:** User was successfully redirected to the checkout step one page.
- **Pass/Fail:** Pass

6. Risk Assessment

Risk Identified	Impact: What happens if it fails?	Severity / Probability	Mitigation Strategy
Problem User UI Breaks: Form elements, buttons, or product images failing on problem_user profile.	Users may experience broken layouts, unclickable buttons, or incorrect product images, leading to a poor user experience.	High / High	Implement explicit assertion timeouts and visual validation checks in automated scripts.
Performance Latency: Response delays under performance_glitch_user profile.	Page loads take too long, causing test script timeouts or user frustration due to freezing screens.	Medium / High	Utilize Selenium Explicit Waits (WebDriverWait) instead of fixed/implicit sleeps to handle slow loading.
Cart Session State Loss: Cart items dropping or resetting during step redirections in checkout.	Users lose selected items when moving from the cart to the checkout page, preventing order completion.	High / Low	Execute end-to-end regression test suites across multiple browsers (Chrome, Firefox) to ensure session persistence.